

Indian Institute of Technology Gandhinagar

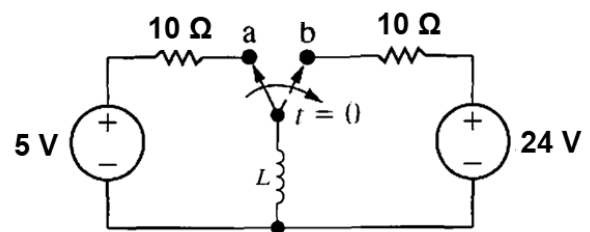
ES116 Principles and Applications of Electrical Engineering

Sem II, 2025-2026

Tutorial-4 (Response of Electrical Circuits with Inductors/Capacitors and Resistors)

In this tutorial, you will explore how Resistors (Energy Dissipators), Inductors (Current Stabilizers), and Capacitors (Voltage Stabilizers) behave under different conditions. The goal is to move beyond abstract equations and understand the physical personality of each component.

Q1. You are analyzing a fuel injector solenoid. It can be modeled as a series R-L circuit ($R = 10\ \Omega$, $L = 100\text{ mH}$). Instead of a simple on/off switch, the circuit is controlled by a 2-Position Switch, as shown in the figure.

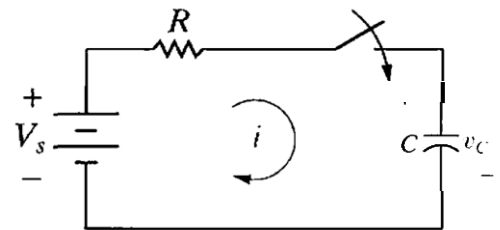


The switch has been in Position A for a long time (Steady State 1). At $t=0$, the switch instantly flips to Position B.

- Calculate the current flowing just before the switch flips, $i(0^-)$. What is the current $i(0^+)$ the exact instant the 24 V source is connected? Can the current jump to 2.4 A immediately?
- At $t=0^+$, calculate the voltage across the inductor $v_L(0^+)$ at this exact moment. Can the inductor voltage be *almost* as high as the source?
- Write the mathematical expression for $i(t)$ for $t > 0$. (*Hint: Use the General Solution $i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau}$).*
- However, the fuel injector is a mechanical valve held shut by a stiff spring. The magnetic force is only strong enough to pull the valve open once the current exceeds 1.5A. Use your equation to calculate the exact time t_{open} when the current crosses the 1.5A threshold.
- The engine control unit (ECU) needs this injector to open within 3 ms of the command signal to ensure fuel enters the cylinder at the correct moment for combustion. Does your circuit meet this specification, or is the inductor too slow?

Q2. You are designing the Burst Mode for a professional camera flash. The given circuit models the phase 2, but it is important to know the history of the circuit (Phase 1):

- Phase 1 ($t < 0$): When the flash fires, the voltage drops rapidly. However, the Xenon tube stops conducting (extinguishes) when the voltage drops to 50 V. It doesn't go to zero! ($v_c(0) = 50\text{ V}$.)
- Phase 2 ($t = 0^+$): Immediately after the flash cuts off, the charging circuit ($V_s = 300\text{ V}$, $R = 1\text{ k}\Omega$) activates to refill the capacitor ($C = 1\text{ mF}$).



A photographer takes a picture at $t < 0$. The flash fires and extinguishes, leaving the capacitor with a Residual Voltage of 50 V.

- A. Calculate the initial charging current $i(0^+)$ the moment the recharge begins. Does having this 50 V head start make the initial current higher or lower compared to charging from 0V?
- B. Write the mathematical expression for the voltage $v_c(t)$ for $t > 0$. (*Hint: Use the General Solution $v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$.*)
- C. The photographer is shooting a fast sports event and presses the shutter button again just 0.8 s after the first shot. Calculate the voltage across the capacitor at exactly $t=0.8$ s.
- D. For a "Good Exposure," the flash needs to deliver at least 15 J of energy. Did the capacitor accumulate enough energy by 0.8s to take a good photo, or will the second picture be underexposed (dark)?

Q3. You are programming the motion control for the knee joint of a humanoid robot. The joint uses a high-performance Brushless Motor. One phase of the motor winding is modeled as a Resistor ($R = 4 \Omega$) and an Inductor ($L = 10$ mH) in series. To hold a static crouched position, the motor driver synthesizes a steady sinusoidal voltage: $v(t) = 24 \sin(400 t)$ V. (*Note: 400 rad/s is the electrical frequency of the drive signal*).

You need to calculate the Steady State Current flowing through the winding to ensure the motor driver chip won't overheat (Limit = 4.0 A).

- A. Write a differential equation for this circuit in the time domain to find $i(t)$ and $v(t)$. Why is this approach considered computationally expensive for a real-time controller?
- B. Transform the problem into the Frequency Domain to solve it using algebra: convert the source $v(t)$ to a Phasor, and calculating the Impedance (Z_{total}), find the current Phasor Magnitude and corresponding $i(t)$.
- C. Based on your Phasor result, does the current exceed the 4.0 A safety limit? Why is it safe to ignore the initial startup transient when analyzing a robot holding a pose for several seconds?

Q4. You are building an analog Noise Filter for the robot's voice recognition microphone. The robot works in a noisy environment (humming servers and hissing steam). You need a circuit that blocks low rumble and high hiss, but passes the human voice clearly. Based on your friend's advice you use a series R-L-C circuit, where $R = 100 \Omega$ representing the Output/ADC input impedance), $L = 10$ mH, and $C = 1$ uF. Input is a microphone signal V_{in} containing many different frequencies.

- A. At very low frequencies ($\omega \rightarrow 0$), how does the capacitor act like (a short circuit or an open circuit)? Will the low-frequency rumble get through to the resistor?
- B. At very high frequencies ($\omega \rightarrow \infty$), how does the inductor act like (a short circuit or an open circuit)? Will the high-frequency hiss get through? Across which component would you take the output?

- C. Can the Inductor and Capacitor Reactance's effectively cancel each other out at specific frequency? If yes, calculate that frequency.
- D. Does the circuit allow maximum or minimum current at this frequency?
- E. If you wanted to make the filter sharper (listening *only* to a specific tone and rejecting everything else aggressively), would you increase or decrease the Resistance R?

More Practice Problems

- Irwin J.D. and Nelms R.M., Basic Engineering Circuit Analysis.
- Nilsson J., S.-Riedel, Electric-Circuits, 9th-ed, Prentice-Hall-2011.